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IOT-BASED GAIT DISORDER DETECTION SYSTEM

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ABSTRACT

Our system leverages IoT technology to detect gait disorders by continuously tracking essential health indicators and walking patterns through lightweight wearable sensors. The system is built around the ESP32 microcontroller and integrates components such as the MAX30100 pulse oximeter, a temperature sensor, and the ADXL345 accelerometer, which is positioned on the leg to track gait movements. An OLED screen provides immediate, on-device display of sensor readings, while a buzzer delivers real-time alerts when irregularities are detected. The setup monitors heart rate, blood oxygen saturation (SpO_2), body temperature, and gait dynamics. All sensor data is both visualized on the OLED screen and wirelessly uploaded to the ThingSpeak cloud platform through Wi-Fi for remote access. A web-based application, developed using the Flask framework, pulls this data from the cloud, compares it against baseline normal values, and presents it visually for analysis. The system identifies whether the current readings fall within normal ranges or exhibit abnormalities, and it flags potential gait-related disorders. When any health metric or gait pattern crosses pre-set safe limits, the buzzer activates immediately to notify caregivers. This cost-efficient, scalable system is designed to improve patient safety and support early detection of health and mobility concerns through real-time, remote tracking..

Keywords: IOT , WI-FI , Microcontroller, OLED, cloud server

I. INTRODUCTION

An IoT-Based Gait Disorder Detection System is a smart healthcare solution designed to continuously monitor vital health metrics and walking patterns using wearable sensors. This system plays a crucial role in tracking patient recovery, especially for individuals affected by neurological disorders, surgeries, or injuries that impact their gait and mobility. At the core of the system is the ESP32 microcontroller, which integrates multiple sensors including a pulse oximeter (MAX30100) to monitor heart rate and SpO_2 levels, a temperature sensor for body temperature, and an ADXL345 accelerometer mounted on the leg to capture gait-related movement. These wearable sensors collect real-time data, enabling accurate tracking of the patient's physiological condition and mobility status. The gathered sensor information is shown on an OLED display in real time and wirelessly sent to the ThingSpeak cloud server using Wi-Fi. A web application built using Flask retrieves this data, compares it with a predefined normal dataset, and determines whether the current readings are normal or indicate abnormalities. The gait data is also analyzed to detect potential disorders based on unusual leg movement patterns. An additional safety feature is provided through a buzzer, which activates to alert caregivers when any reading crosses a dangerous threshold. This system offers an efficient, low-cost, and non-invasive method for remote monitoring, allowing healthcare professionals to assess recovery progress from a distance, make timely interventions, and reduce hospital readmission rates. By supporting early detection and data-driven decisions, the system enhances patient care and rehabilitation outcomes.

II. METHODOLOGY

A. Hardware Integration: The system centers around the ESP32 microcontroller, which is ideal for wearable health monitoring due to its low power consumption and built-in Wi-Fi. The following sensors are interfaced with ESP32:

- **MAX30100 sensor** for measuring heart rate and blood oxygen (SpO_2) levels.
- Utilizes the DS18B20 sensor to measure body temperature.

- **ADXL345 accelerometer** mounted on the leg for gait analysis.
- An **OLED display** provides local real-time visual feedback.
- A **buzzer** acts as an alert mechanism in case of abnormal sensor readings. The system operates on battery power, allowing for portable and uninterrupted monitoring

B. Data Collection : The sensors continuously collect real-time physiological and motion data:

- **MAX30100** captures pulse rate and oxygen saturation.
- **Temperature sensor** monitors body temperature.
- **ADXL345** records leg movements to assess step count, balance, and possible gait disorders. All sensor data is collected by the ESP32, which acts as the central processing unit of the wearable device.

C. Data Processing: The ESP32 processes raw sensor data before transmission.

- For **gait analysis**, basic filtering techniques (like low-pass filters) are applied to reduce noise in ADXL345 data.
- Step count and irregular movement patterns are computed using threshold-based algorithms.
- Abnormalities in the MAX30100 and temperature sensor data are continuously monitored.
- All values are formatted and prepared for cloud upload and local display.

D. Data Transmission: The ESP32 uses its built-in Wi-Fi to transmit the processed data to the **ThingSpeak cloud platform**.

- Data is sent at regular intervals using HTTP protocol.
- ThingSpeak serves as a protected cloud storage platform, facilitating live data uploads and remote access.

E. Visualization and Monitoring A**Flask-based web application** is developed to fetch and display live data from ThingSpeak.

- The interface shows parameters such as **BPM**, **SpO_2**, **temperature**, and **gait movement status**.
- The system compares the live values to a pre-defined normal dataset to determine whether readings are within safe limits.
- If any abnormal condition is detected (e.g., low SpO_2 or irregular gait), a **buzzer alert** is triggered, and the web interface flags the condition for caregivers or doctors.

III. SYSTEM IMPLEMENTATION

A. WORKING PRINCIPLE OF ESP32 :

The project employs an ESP32 microcontroller as the primary processing unit. The ESP32 is chosen for its built-in Wi-Fi module, multiple GPIO pins, and efficient data-handling capabilities. It supports various communication protocols like I2C, SPI, and UART, making it versatile for integrating sensors and peripherals. Its Wi-Fi functionality enables seamless data transmission to the cloud, ensuring real-time monitoring and storage. Additionally, its low power consumption makes it suitable for wearable applications.

B. ACCELEROMETER :

An accelerometer, such as the MPU6050, is used to capture motion data by measuring acceleration along three axes (X, Y, Z). This sensor detects gait-related parameters, including step count, stride length, and walking speed. The raw data is processed to remove noise and derive meaningful metrics using low-pass filtering and threshold-based algorithms. The accelerometer is essential for detecting abnormalities in gait, helping to assess how well a patient is recovering.

C. OLED DISPLAY :

An OLED display is integrated into the system to provide instant feedback to the user. This compact and power-saving screen presents key information, including the number of steps taken, Wi-Fi connection status, and important system notifications. It ensures that the system remains user-friendly, allowing patients to check their progress directly on the device without needing additional tools.

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D. WI-FI MODULE :

The built-in Wi-Fi module of the ESP32 is utilized to connect the device to a cloud server. The Wi-Fi capability facilitates real-time data transmission, allowing seamless communication between the device and the cloud platform. This ensures that data is centralized and accessible for remote monitoring by healthcare providers.

E. WEB APP :

A web application provides the interface for caregivers and doctors to access the collected data. The app presents recovery data through visual tools such as charts and graphs, highlighting patterns in metrics like step count and walking pace over a period of time. It also generates alerts in case of abnormal gait patterns, allowing healthcare professionals to intervene promptly. This helps the system facilitate prompt and efficient monitoring of the patient's recovery process.

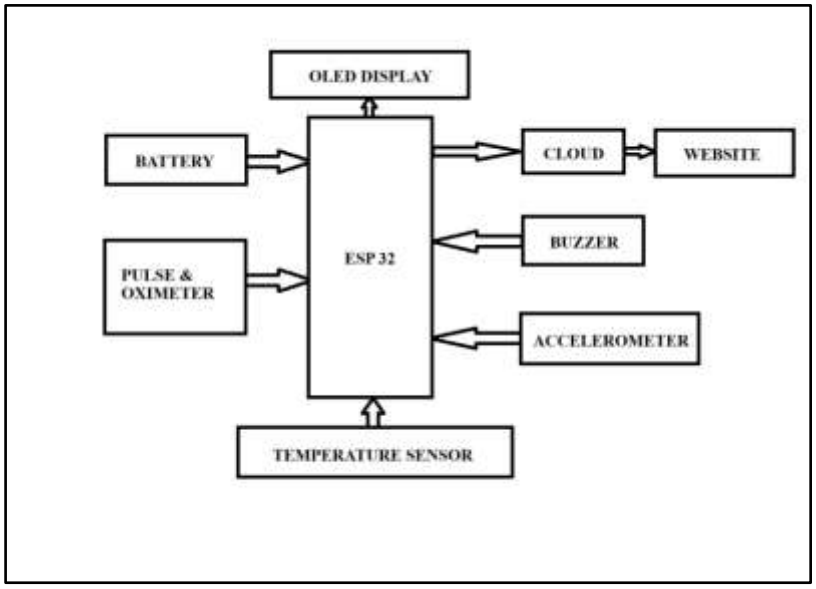

Figure 1: Block Diagram of proposed system

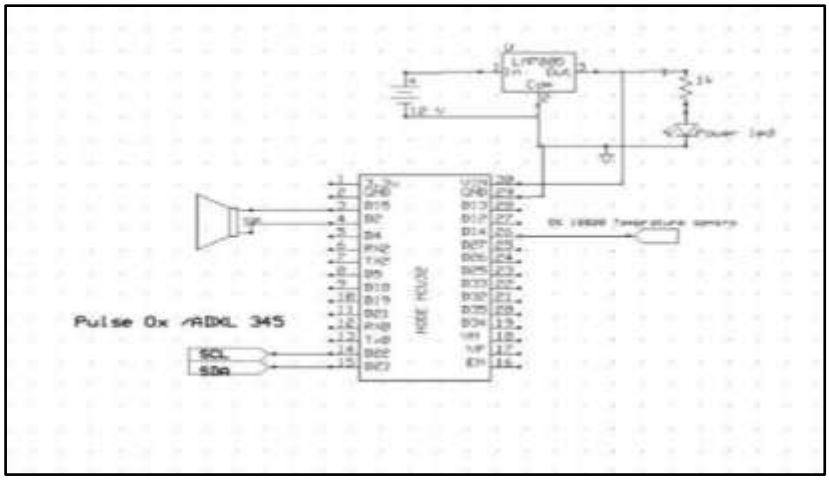

Figure 2 : Circuit Diagram

IV. IMPLEMENTATION DETAILS

The IoT-Based Gait Disorder Detection System is a wearable solution designed for real-time monitoring of vital health parameters and gait activity. It integrates various biomedical and motion sensors with the ESP32 microcontroller to collect, process, and transmit data. The data is displayed locally on an OLED and uploaded to a cloud platform (ThingSpeak) via Wi-Fi. A Flask-based web application provides a user-friendly interface for remote monitoring and analysis.

Hardware Components Used:

Microcontroller Unit (MCU): ESP32

Acts as the central processing and communication hub.Equipped with integrated Wi-Fi and Bluetooth modules to enable seamless wireless communication. Controls sensors, display, and alert system. Performs initial data filtering and formatting before transmission.

Sensors Used :

Pulse & SpO_2 Sensor (MAX30100):

Measures heart rate and blood oxygen saturation (SpO_2). Worn on the finger or wrist. Helps detect physical stress and abnormal vitals.

Temperature Sensor (LM35):

Monitors patient's body temperature.Crucial for identifying fever or signs of infection after surgery.

Accelerometer (ADXL345):

Monitors walking patterns such as speed, number of steps, and stability, with the sensor positioned on the leg to ensure precise motion tracking.

Wireless Communication :

Wi-Fi via ESP32:

Data is transmitted live to the ThingSpeak cloud service, facilitating monitoring from any internet-enabled location.

Power Supply :

Li-ion Rechargeable Battery (3.7V, 2000mAh):

Powers the wearable device.Incorporates an LM7805 voltage regulator to provide a consistent 5V power output. Ensures portability and continuous operation.

Display and Alerts :

OLED Display (0.96" I2C):

Displays BPM, SpO_2, temperature, and gait info locally. Delivers immediate alerts to both patients and caregivers.

Buzzer:

Activates when abnormal values are detected.Valuable for providing instant notifications during urgent situations.

Sensor Placement :

Accelerometer: Mounted on the leg (ankle/shin) to detect gait movements. **MAX30100**: Worn on fingertip or wrist for vitals monitoring. **Temperature Sensor**: Positioned on the skin (arm or wrist). Components are enclosed in a lightweight, compact case for easy wearability.

Data Processing &Transmission :

Sensor readings are filtered (e.g., low-pass filter) for accuracy. Step detection is performed using threshold-based methods on accelerometer data. Vitals are evaluated against threshold values to trigger alerts. The processed information is sent to ThingSpeak via the HTTP communication protocol.

Power Management :

The system uses efficient components to minimize power usage. Sleep modes are implemented in the ESP32 to conserve battery when idle. Battery status is continuously checked to maintain system dependability.

Software Implementation:

Embedded Programming (ESP32)

C/C++ (Arduino IDE):

Reads sensor values. Applies filters and logic for abnormal condition detection. Sends data via Wi-Fi to ThingSpeak.

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Cloud Platform: ThingSpeak :

Stores data received from ESP32. Allows visualization through live graphs. Public or private dashboard can be created.

Web Interface: Flask Application :

Developed in Python using Flask framework. Fetches live data from ThingSpeak API. Displays parameters with real-time status: Normal / Abnormal. Highlights alerts and anomalies for faster action.

Data Comparison Logic :

Flask app compares live sensor data with predefined normal ranges. Displays results as "Normal" or "Abnormal" using colored indicators. Supports trend analysis using historical data from ThingSpeak.

V. RESULTS AND DISCUSSION

Hardware Result :

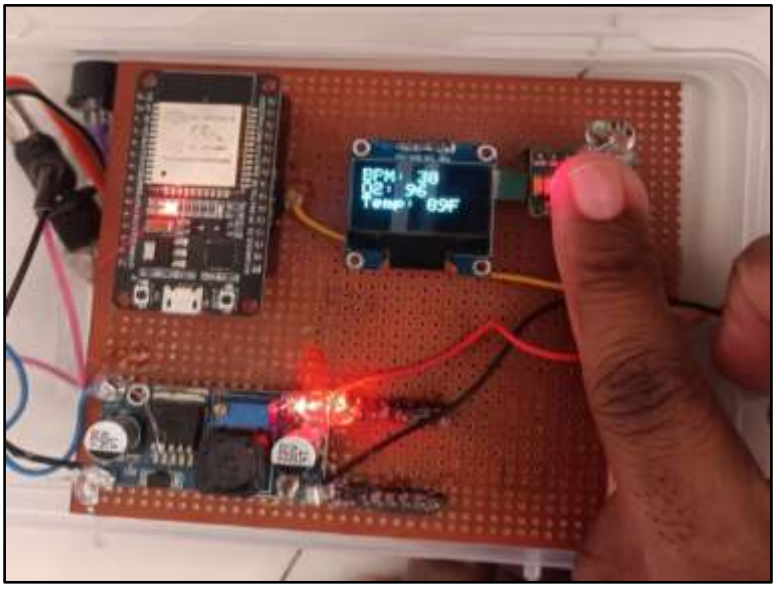

Figure 3. Hardware Implementation

Cloud Result :

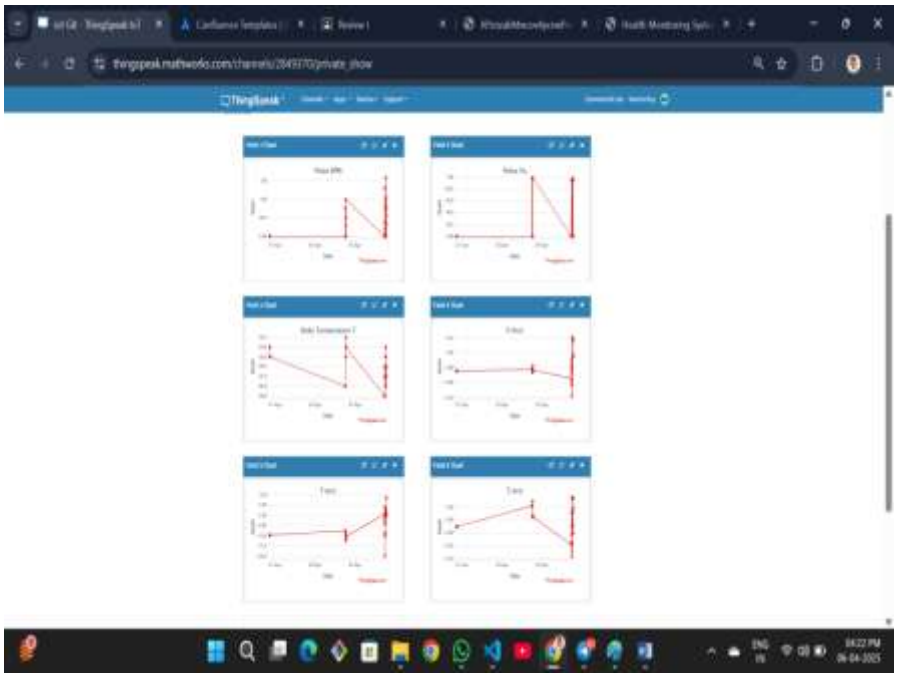

Figure 4. Results on ThingSpeak

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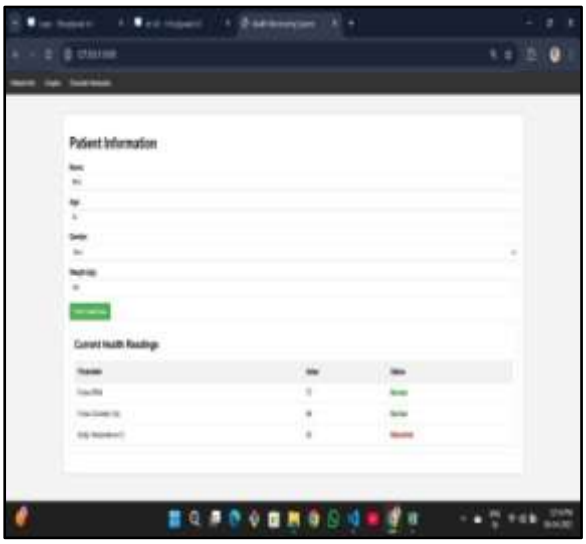
Software Result :



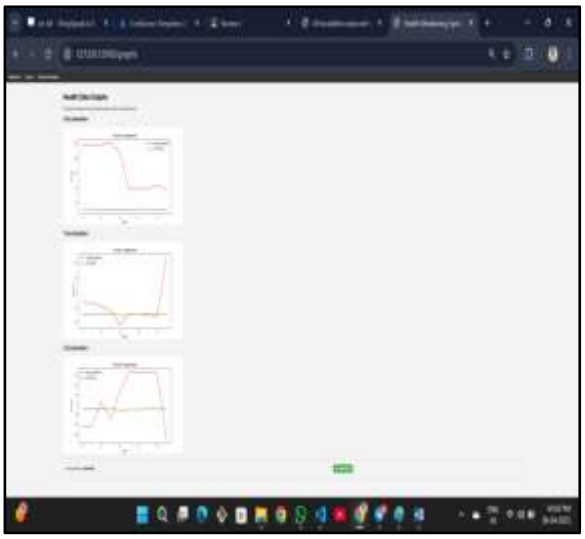


Figure 5. Website Patient Information Page**Figure 6.**Patient's Health Data Graphs

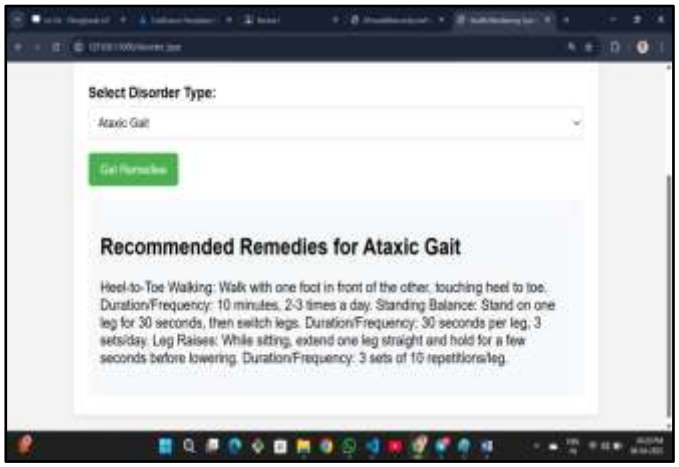


Figure 7. Disorder Remedies Page

V. CONCLUSION

The **IoT-Based Gait Disorder Detection System** successfully integrates wearable sensors and real-time data processing to monitor vital signs and gait-related parameters. Utilizing the ESP32 microcontroller, the system efficiently collects data from sensors such as MAX30100 (for pulse and SpO2), temperature sensors, and the ADXL345 accelerometer. The collected data is displayed on an OLED and transmitted to ThingSpeak, where a Flask-based web interface analyzes and visualizes the data. This enables early detection of abnormal conditions and supports medical professionals in tracking patient recovery. The system also improves patient safety by providing buzzer alerts for abnormal readings and enables remote monitoring, reducing the need for frequent hospital visits. The portability and modular design make it suitable for use in hospitals, rehabilitation centers, elderly care, and home-based recovery.

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